Chebyshev–Lucas Error Correction in Decimated Transfer–Matrix Chains: A Simple, First–Principles Framework with Robust Decoding

VR & AI

August 24, 2025

Abstract

We show that any one–dimensional medium governed by a constant 2×2 unit–cell transfer matrix M admits a *universal tri–site recurrence* when stroboscopically sampled every l sites:

$$x_{k+2l} \;-\; V_l\, x_{k+l} \;+\; Q^l\, x_k \;=\; 0, \qquad V_l = \mathrm{tr}(M^l), \;\; Q = \det M, \;\; Q^l = \det(M^l) = (\det M)^l.$$

This identity follows directly from Cayley–Hamilton applied to the l–step monodromy M^l. We recast it as a *convolutional constraint* for *event–coded* measurements and obtain a *Chebyshev–Lucas checksum* enabling noise–robust decoding via a finite–state trellis (Viterbi/dynamic programming). We give rules for choosing the alphabet and window, prove existence of a unique maximum–likelihood (ML) decoded sequence on a finite window, derive logarithmic bounds on the occupied support radius under repeated injection, and extend the construction to weakly coupled multi–track systems by diagonalizing transverse modes. We also clarify when "symmetric regrouping" into Dickson polynomials applies, provide a counterexample showing the checksum alone is not injective (hence normalization/ML decoding matters), and include concrete validation metrics.

Contents

1 From unit cell to decimated tri–site identity **2**
- 1.1 First principles: second order $\Rightarrow 2 \times 2$ transfer . . . 2
- 1.2 Cayley–Hamilton on the l–step monodromy . . . 2

2 Events, anchor, and the Chebyshev–Lucas checksum **2**
- 2.1 Eventization (operational) . . . 2
- 2.2 Invariant and checksum; when "symmetric regrouping" applies . . . 2

3 Decoding on a window: trellis, design rules, and streaming **3**
- 3.1 Noise tolerance (explicit constants) . . . 4

4 Support radius under repeated anchor injection **4**

5 Uniqueness: what is and is not determined **4**

6 Cross–track coupling: per–mode generalization and norms **5**

7 Choosing the stride l (natural origins) **5**

8 Experimental protocols and validation metrics **5**

9 Related work (selected) 5

A Additional proofs 6

1 From unit cell to decimated tri–site identity

1.1 First principles: second order $\Rightarrow 2 \times 2$ transfer

Many one–dimensional media reduce to a scalar second–order recurrence after eliminating a conjugate variable (momentum/flux):

$$x_{k+1} \;=\; P\,x_k \;-\; Q\,x_{k-1}, \qquad P, Q \in \mathbb{C},\; Q \neq 0. \tag{1}$$

Collecting into a first–order map,

$$\begin{pmatrix} x_{k+1} \\ x_k \end{pmatrix} \;=\; M \begin{pmatrix} x_k \\ x_{k-1} \end{pmatrix}, \qquad M = \begin{pmatrix} P & -Q \\ 1 & 0 \end{pmatrix}, \quad \mathrm{tr} M = P,\; \det M = Q. \tag{2}$$

1.2 Cayley–Hamilton on the l–step monodromy

Proposition 1.1 (Decimated recurrence)**.** *For any $l \in \mathbb{N}$,*

$$M^{2l} - \mathrm{tr}(M^l)\, M^l + \det(M^l)\, I \;=\; 0. \tag{3}$$

Applying (3) to $\binom{x_k}{x_{k-1}}$ and taking the first component yields

$$x_{k+2l} - V_l\, x_{k+l} + Q^l\, x_k = 0, \qquad V_l := \mathrm{tr}(M^l),\; Q^l := \det(M^l) = (\det M)^l. \tag{4}$$

Remark 1.2 (Lucas/Dickson polynomials). If $\lambda_\pm$ are eigenvalues of M, then $V_l = \lambda_+^l + \lambda_-^l$ obeys $V_{n+1} = PV_n - QV_{n-1}$ with $V_0 = 2$, $V_1 = P$.

2 Events, anchor, and the Chebyshev–Lucas checksum

2.1 Eventization (operational)

Fix measurement cadence Δt. For each site t let $E_t[n]$ be integrated power/flux in window n. Choose a quantum/threshold $\varepsilon > 0$ and set

$$\delta N_t[n] \;:=\; \left\lfloor \frac{E_t[n] - E_t[n-1]}{\varepsilon} \right\rfloor \in \mathbb{Z}_{\geq 0}, \qquad T_s := \sum_n \delta N_s[n]. \tag{5}$$

An *event* at (t, n) is $\delta N_t[n] > 0$. The *anchor* experiment restricts events to $s = 0$ (or pre–aligns them by virtual index shifts).

2.2 Invariant and checksum; when "symmetric regrouping" applies

Let r be a root of $z^2 - V_l z + Q^l = 0$ with $|r| \geq 1$ (the other root is Q^l/r). For integer sequences $d = \{d_k\}$ with finite support define

$$I(d) \;=\; \sum_k d_k r^k. \tag{6}$$

Lemma 2.1 (Invariance under a tri–site move)**.** *The local update $(d_{t-1}, d_t, d_{t+1}) \mapsto (d_{t-1} + Q^l,\; d_t - V_l,\; d_{t+1} + 1)$ leaves $I(d)$ unchanged.*

Proof. The change is $-V_l r^t + r^{t+1} + Q^l r^{t-1} = 0$ by the characteristic equation. □

Theorem 2.2 (Generalized checksum)**.** *If d is obtained from an event stream $\{\delta N_s[n]\}$ by any sequence of local tri–site moves, then*

$$\sum_k d_k\, r^k \;=\; \sum_s T_s\, r^s. \tag{7}$$

Proof. Each unit event at site s increases I by r^s. Lemma 2.1 shows I is invariant under subsequent tri–site moves. □

Symmetric regrouping (Dickson form). On a *symmetric window* $[-R, R]$ and when the decoded digits satisfy $d_{-t} = d_t$ (e.g., due to symmetric injection/BC or explicit symmetrization of the window), one may regroup

$$\sum_{k=-R}^{R} d_k r^k \;=\; d_0 + \sum_{j=1}^{R} d_j \left(r^j + (Q^l/r)^j\right) \;=\; d_0 + \sum_{j=1}^{R} d_j\, D_j(V_l, Q^l), \tag{8}$$

where $D_0 = 2$, $D_1 = V_l$, and $D_{j+1} = V_l D_j - Q^l D_{j-1}$ are Dickson polynomials ($D_j(V_l, Q^l) = r^j + (Q^l/r)^j$). Without symmetry, keep the two geometric progressions separately; the invariant (7) still holds.

Example 2.3 (Illustration for $l = 1$). Let $M = \left(\begin{smallmatrix} P & -Q \\ 1 & 0 \end{smallmatrix}\right)$ with $P = 1$, $Q = -1$ so $V_1 = 1$, $Q^1 = -1$, and r solves $r^2 - r - 1 = 0$ ($r = \varphi$). Anchor events $T = 5$ yield decoded digits $d_0 = 5$, others zero: $\sum_k d_k r^k = 5$ matches (7). With $T = 11$, a carry creates $(d_{-1}, d_0, d_{+1}) = (1, 8, 1)$, giving $11 = 8 + 1 \cdot (r + (Q/r)) = 8 + 1 \cdot D_1(V_1, Q^1)$.

3 Decoding on a window: trellis, design rules, and streaming

We observe $m_t = d_t + \eta_t$ on $t \in [-R, R]$.

Definition 3.1 (Decoder (nearest–integer, constrained))**.**

$$\hat{d} \in \arg\min_{d \in \mathcal{D}^{2R+1}} \sum_{t=-R}^{R} w_t\, |d_t - m_t|^2 \quad \text{s.t.} \quad d_{t+1} = V_l d_t - Q^l d_{t-1} \;\; \forall t \in [-R, R-1]. \tag{9}$$

Proposition 3.2 (Finite–state trellis)**.** *States are pairs $(a, b) = (d_{t-1}, d_t) \in \mathcal{D}^2$. A transition $(a, b) \to (b, c)$ is allowed if $c = V_l b - Q^l a$ lies in $\mathcal{D}$ (soft trellis: $c = \mathrm{round}(\cdot)$ with a penalty). Dynamic programming yields the ML path in $O(R\,|\mathcal{D}|^2)$ time.*

Design rules (alphabet and window). Let $r > 1$ be the larger root of $z^2 - V_l z + Q^l = 0$. For nonnegative $\mathcal{D}^+ = \{0, 1, \ldots, A\}$, avoid saturation by

$$A_{\min}(R) \;:=\; \left\lceil \frac{T_{\max}}{1 + \sum_{j=1}^{R} D_j(V_l, Q^l)} \right\rceil, \qquad A \;=\; \max\Big\{ A_{\min}(R),\; \lfloor \sqrt{|V_l|}\, \rfloor \Big\}, \tag{10}$$

and choose

$$R \;=\; \left\lceil \log_r\!\Big(1 + \frac{(r-1)T_{\max}}{A}\Big) \right\rceil - 1 \;+\; m, \qquad m \in \{2, 3\}. \tag{11}$$

Balanced $\mathcal{D}^{\pm} = \{-A, \ldots, A\}$ uses the same rules.

Streaming (guard states). Maintain a sliding window. At each step, freeze the left guard pair (d_{-R-1}, d_{-R}) from the previous solution, prune transitions violating the recurrence at $t = -R$, and run one DP pass over $[-R+1, \ldots, R+1]$. This yields amortized $O(|\mathcal{D}|^2)$ updates per cadence.

3.1 Noise tolerance (explicit constants)

Assume $m_t = d_t + \eta_t$ with i.i.d. $\eta_t \sim \mathcal{N}(0, \sigma^2)$ and d supported on $[-R, R]$. A union bound over trellis paths gives

$$\mathbb{P}(\hat{d} \neq d) \;\leq\; C_R \exp\Big(-\frac{d_{\min}^2}{8\sigma^2}\Big), \qquad C_R \;\leq\; (2R{+}1)\,|\mathcal{D}|^2, \tag{12}$$

where $d_{\min}$ is the minimum Euclidean distance between distinct trellis paths and obeys the conservative estimate

$$d_{\min}^2 \;\gtrsim\; \min\{r^{2R},\ (Q^l/r)^{2R}\}. \tag{13}$$

Thus one admissible operating point is $\sigma_\star \leq \frac{1}{\sqrt{8}} \min\{r^R, (Q^l/r)^R\}$.

4 Support radius under repeated anchor injection

Let $R(T) = \max\{|t| :\ d_t(T) \neq 0\}$ after T anchor events and decoding. Define

$$S_R(V_l, Q^l) \;:=\; 1 + \sum_{j=1}^{R} D_j(V_l, Q^l). \tag{14}$$

Proposition 4.1 (Maximal representable count at radius R)**.** *For $\mathcal{D}^+ = \{0, \ldots, A\}$,*

$$T_{\max}(R) \;=\; A\, S_R(V_l, Q^l). \tag{15}$$

Remark 4.2 (Why S_R starts with 1 (not D_0))*.* The checksum (8) uses d_0 with coefficient 1 (the term r^0), whereas $D_0 = 2$ corresponds to the symmetric combination $r^0 + (Q^l/r)^0 = 2$. Maximizing T at fixed radius sets all digits $(d_0, d_1, \ldots, d_R)$ to A, yielding $T_{\max} = A\,(1{+}\sum_{j=1}^{R} D_j)$, consistent with (14).

Using $D_j(V_l, Q^l) = r^j + (Q^l/r)^j$ gives

$$\frac{r^{R+1} - 1}{r - 1} \;\leq\; S_R(V_l, Q^l) \;\leq\; \frac{2(r^{R+1} - 1)}{r - 1}. \tag{16}$$

Theorem 4.3 (Bounds and scaling)**.**

$$\Big\lceil \log_r\Big(1 + \frac{(r-1)T}{2A}\Big)\Big\rceil - 1 \;\leq\; R(T) \;\leq\; \Big\lceil \log_r\Big(1 + \frac{(r-1)T}{A}\Big)\Big\rceil - 1. \tag{17}$$

Hence $R(T) = \Theta(\log_r T)$ for $r > 1$.

Remark 4.4 (Marginal $r \approx 1$)*.* If $|V_l| \approx 2\sqrt{Q^l}$ then $r \approx 1$ and confinement degrades; choose the stride l (supercell/Floquet) to enforce $|V_l| \geq 2\sqrt{Q^l} + \varepsilon$.

5 Uniqueness: what is and is not determined

Checksum is not injective (counterexample). Let Δ be zero everywhere except at $(t{-}1, t, t{+}1)$ with values $(Q^l,\ -V_l,\ 1)$. Then $\sum_k \Delta_k r^k = 0$ by the characteristic equation. Thus the checksum *alone* does not determine d.

Anchor case: uniqueness via normalization/ML on a window. On a finite window with a fixed alphabet and the recurrence constraint (9), the ML decoder is a strictly convex quadratic over a finite trellis; ties are broken deterministically by the DP backtrack, hence the decoded $\hat{d}$ is *unique*. This matches the canonical normal form selected by the local normalization calculus.

Multi–site events. For general $\{T_s\}$, the checksum (7) fixes $\sum_k d_k r^k$ but, by the counterexample, does not *by itself* fix d. Uniqueness is recovered by the same ML principle on a window (fixed alphabet, recurrence constraints) and, if desired, symmetric windows (which eliminate sign–flip ambiguities by construction). When per–site totals are known, one may pre–align to the anchor frame and decode per shift; the superposition is then validated by the generalized checksum.

6 Cross–track coupling: per–mode generalization and norms

For a multi–track chain with nearest–track coupling, transverse Fourier diagonalization yields per–mode transfers $M(\theta)$. We use the operator norm $\|\cdot\|_2$ unless otherwise stated.

Proposition 6.1 (Per–mode decimation)**.** *Each mode θ obeys*

$$x_{k+2l}^{(\theta)} - V_l(\theta)\, x_{k+l}^{(\theta)} + Q^l(\theta)\, x_k^{(\theta)} = 0, \qquad V_l(\theta) = \mathrm{tr}(M(\theta)^l), \quad Q^l(\theta) = \det(M(\theta)^l). \tag{18}$$

Quantitative robustness. If $M(\theta) \mapsto M(\theta) + E(\theta)$ with perturbation E, then

$$\left|\mathrm{tr}((M+E)^l) - \mathrm{tr}(M^l)\right| \;\le\; l\, \|M\|_2^{l-1}\, \|E\|_2 + O(\|E\|_2^2). \tag{19}$$

We call coupling *weak at level* δ if $l\, \|M\|_2^{l-1}\, \|E\|_2 \le \delta\, |\mathrm{tr}(M^l)|$ uniformly in θ; under this condition the per–mode checksum is accurate to $O(\delta)$.

7 Choosing the stride l (natural origins)

- **Supercell:** l equals the microperiod (geometry).
- **Floquet locking:** with phase advance θ per site, pick l s.t. $\theta/\pi = p/l$ rational, making $V_l = 2\cos(l\theta)$ typically small integers.
- **Topology:** on a ring, one round–trip per residue class $\Rightarrow$ decimation by l; APBC (π–flux) flips the global holonomy.

8 Experimental protocols and validation metrics

P0–P4 as in the main version, with added plots: checksum accuracy vs. $\sigma/\sigma_\star$, decoder success rate vs. $\sigma/\sigma_\star$, and $R(T)$ scaling with the theoretical bounds.

9 Related work (selected)

- **Transfer matrices and decimation:** G. Economou, *Green's Functions in Quantum Physics*, 3rd ed., Springer, 2006.
- **Chebyshev polynomials:** T. J. Rivlin, *Chebyshev Polynomials*, 2nd ed., Wiley, 1990.
- **Dickson polynomials:** R. Lidl and H. Niederreiter, *Finite Fields*, Cambridge Univ. Press, 1997 (see chs. on Dickson polynomials).
- **Convolutional codes and trellises:** A. J. Viterbi, "Error bounds for convolutional codes and an asymptotically optimum decoding algorithm," *IEEE Trans. Inf. Theory*, 1967; G. D. Forney Jr., "Convolutional codes I: Algebraic structure," *IEEE Trans. Inf. Theory*, 1970.

Contrast to classical trellises. Classical trellises are engineered algebraically; here the parity constraint (kernel) and alphabet are *medium–determined* via (V_l, Q^l) from the unit cell and the chosen stride l (supercell/Floquet/topology). This enables error–checked, event–coded metrology without explicit coding hardware.

Acknowledgments

We thank the transfer–matrix, special polynomial, and coding theory communities for foundational ideas.

A Additional proofs

Proof of Proposition 1.1. Cayley–Hamilton on M^l gives $M^{2l} - (\mathrm{tr} M^l) M^l + (\det M^l) I = 0$. Acting on $\binom{x_k}{x_{k-1}}$ and taking the first component yields (4). □

Lemma 2.1. Immediate from $r^2 - V_l r + Q^l = 0$. □